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Vierkantswortels van  $20i + 21$

$$20i + 21 = (x + yi)^2$$

$$\begin{cases} x^2 - y^2 &= 21 \\ 2xy &= 20 \end{cases}$$

Tweede vergelijking:

$$2xy = 20 \Rightarrow xy = 10 \Rightarrow x^2(-y^2) = -100$$

$$\begin{cases} x^2 - y^2 &= 21 \\ x^2(-y^2) &= -100 \end{cases}$$

$$\lambda^2 - 21\lambda - 100 = 0$$

$$D = 21^2 - 4 \cdot (-100) = 841$$

$$\lambda_{1,2} = \frac{21 \pm 29}{2} \Rightarrow \lambda_1 = 25, \quad \lambda_2 = -4$$

$$x^2 = 25, \quad (-y^2) = -4$$

$$x = 5 \quad x = -5, \quad y = 2 \quad y = -2$$

De vierkantswortels zijn dus:  $5 + 2i$   $-5 - 2i$

## References